The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer as an anonymized observation runtime: multi-band evidence → BandPIT/WMI → a point-in-time (PIT) *world bundle*, with hard refusal when support is thin. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. ≲0.23 under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: Compiled thin-abstain is 1.0 (100-day and full OOS n=2000) while Ungated soft judgment is only ≈0.68 / ≈0.56; Raw over-trades. Secondary—handoff: vintaged tilts beat momentum under a no-LLM rule (ΔCE=0.334, p=0.034; LOBO content share 1.0). On this panel max WMI ≈ 0.093<0.2, so production Compiled never opens—a valve stress test. **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation → decision → execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

Failure modes. No feed → refuse (safe, useless as a data end). Thin price fragment → trade and lose (unsafe for any downstream stack). Disclosed but ungated world → read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

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Table 1: End-to-end quant stack; this paper owns only the data end.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

What we build. An anonymized runtime (Section 4): multi-band evidence → BandPIT/WMI → JSON world bundle for GPT /DeepSeek /GLM /Gemin class models, served through a read API a decision layer can call. "World model" means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

How we evaluate. **RQ1 (primary—seeing):** Are ready /missing /tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (`mom5`), Blind. **RQ3 (secondary):** Can a transparent no-LLM rule hand off from the bundle (production refuse vs. nonempty tilts)?

Contributions. (1) Data-end semantics: epistemic observations, compilation Π_t, world-quality index, and seeing vs. gating with a lag–missingness bound. (2) Anonymized PIT compile path: collectors→BandPIT→s surface→world bundles; frozen four-arm harness. (3) Live + handoff validation: grounding ≈0.97 with sufficiency dissociation; thin-abstain 1.0 at 100-day and full-OOS scale; no-LLM handoff (production refuse vs. nonempty tilts, ΔCE=0.334; LOBO content share 1.0). Estimand: data-end legibility, valve reliability, and handoff usability.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream at the data end of the stack. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act. Our estimand is therefore orthogonal to memory routers and debate judges: without a typed PIT world, those stacks consume whatever fragment the tool dump provides.

Selective prediction and PIT.. Abstention can be optimal under noise [4, 5]; we attach refusal to *world* quality $q(W_t)$ via a runtime field, not classifier confidence alone (Prop. 3.8). Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness (Prop. 3.6). Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation-Layer Semantics

This section states the data-end theory as *interface semantics*: what counts as an observation, how compilation yields a world, and when quality forces refusal. It is not a pricing theorem and not a generative dynamics model. Systems realization is Section 4; empirical tests are Section 5.

Definition 3.1 (Epistemic observation). $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views into a consumer object.

Definition 3.3 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$. W_t is the handoff object to decision.

Definition 3.4 (World-quality index). Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty in $[0, 1]$; ACWMI multiplies a regime weight. The Compiled interface exposes `should_ai_abstain`$=\mathbf{1}\{q(W_t) < q^\star\}$ and `thin_world` (production $q^\star = 0.2$ for WMI).

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t) < q^\star$. Tilts without a gate are a dump; a gate without a readable bundle is only a switch. The two interfaces are separable: Ungated keeps W_t and numeric $q(W_t)$ but removes the hard boolean.

Assumption 3.5 (Bounded lag and increments). Band j has max observation lag Δ_j; latent increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 3.6 (Lag–missingness reconstruction bound). *Under Assumption 1, any compiler that uses only in-clock observations satisfies*

$$\|\widetilde{S}_t - S_t\| \;\le\; C_{\text{del}}\bar{\delta}\sum_j \Delta_j + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t,$$

where m_t is disclosed missing mass and ε_t collects measurement noise. Operationally, BandPIT maps age to ready/limited/missing so m_t and lag terms are readable in W_t; compilation cannot erase delay, but it can refuse to pretend the bound is zero.

Proposition 3.7 (Compilation ≠ dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty, tightening rather than loosening Prop. 3.6.*

Proposition 3.8 (World-conditional abstention). *Let $c_{\text{act}}(W_t)$ be the expected cost of non-abstain actions and c_{abs} the abstain cost. If $c_{\text{act}}(W_t) > c_{\text{abs}}$ and c_{act} is nonincreasing in $q(W_t)$, the Bayes action is abstain on a lower set $\{W : q(W) < q^\star\}$ [4]. A hard runtime flag implements that set; soft text does not.*

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \to$ WMI; abstain flag; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / availability reads
Consumer harness	Four-arm protocol; temp. 0

Proposition 3.9 (LOBO content vs. gating). *Deleting band j changes a downstream value functional through a content channel and a gating channel. Under an ungated rule the gating channel is shut, so the value gap identifies nonempty fields.*

Theory → protocol. Prop. 3.6–3.7 ⇒ RQ1 (can the consumer recover disclosed readiness/tilts?). Prop. 3.8 ⇒ RQ2 (hard flag vs. soft judgment on the same W_t). Prop. 3.9 ⇒ RQ3 (no-LLM handoff / LOBO content share).

Remark 3.10 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface—the data end of the stack.*

4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path (Figure 1, Table 2). Decision and execution sit downstream; this paper does not claim them. Paths and product names are omitted for double-blind review.

4.1 Layered design

1. **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
2. **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
3. **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty/missingness gates implement Π_t; WMI/ACWMI and availability shocks O_t (band outage events for quasi-exogenous levers) are first-class.
4. **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle and decision-bundle reads, health exposing `should_ai_abstain`, a paper world-model time slice, and availability-shock queries—the machine-readable handoff a decision layer would call.
5. **Consumer layer.** Frozen-prompt LLM/rule agents call an OpenAI-compatible adapter at temperature 0 under four arms.

4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of ≈3990 asset-days (2025-07 to 2026-08): exchange ready share ≈0.80, macro

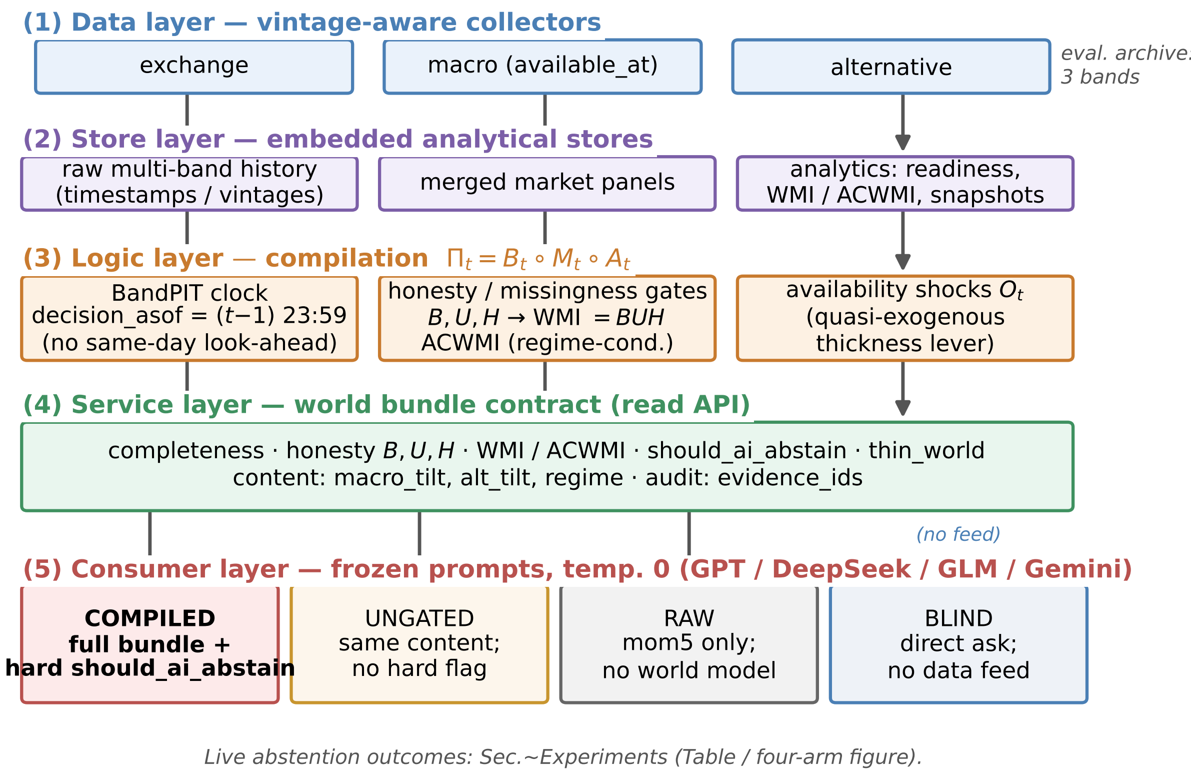

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t, evidence IDs

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

1.0, alternative 1.0. WMI is computed over the full schema, so max WMI ≈ 0.093 even on 3/3-ready days.

4.3 PIT clock and freshness

For calendar day t, BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the bundle records `decision_asof`. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.6). Breadth, stability, and honesty aggregate into $\text{WMI}_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The valve sets `should_ai_abstain` when WMI < 0.2 and exposes `thin_world`. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: WMI stays below the production gate throughout; ACWMI can look healthier under regime weight, so the runtime exports both and lets the configured index mode drive abstain. Table 5: production gate 0.2 never opens on this panel; WMI ≥ 0.05 selects the band-thick subset (≈61% OOS).

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and `evidence_ids`. Listing 1 shows an exchange-gap day; Listing 2 shows a 3/3-ready day that still carries WMI ≈ 0.093 and `should_ai_abstain=true`—the bundle discloses quality rather than inventing coverage.

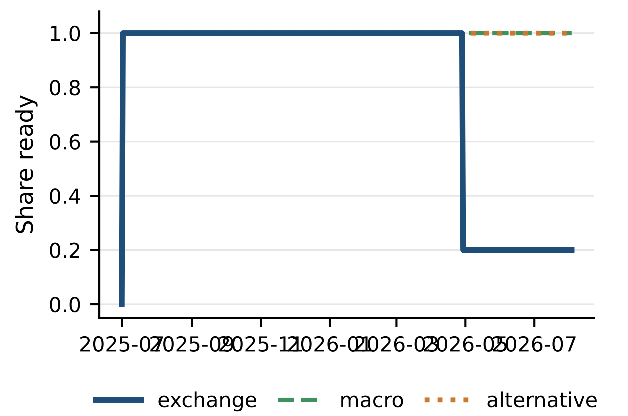

Figure 2: PIT readiness: what is visible in the archive over time.

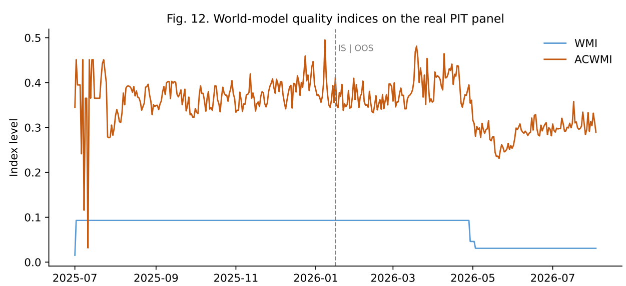

Figure 3: World-quality paths: WMI stays under the 0.2 production gate; ACWMI is regime-weighted.

Table 5: WMI gate vs. open share (OOS). Production never opens on this panel.

Gate	Open share	Note
0.2 (production)	0.00	max WMI ≈0.093
0.05 (counterfactual)	0.61	≡ band-thick
0.01	1.00	near-floor open

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H; main-view / stale gates
Quality	WMI, ACWMI, abstain flag, `thin_world`
Content	macro_tilt, alt_tilt, mom5
Audit	`evidence_ids`; EAR required

Listing 1: Exchange-gap bundle (Compiled).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1},
 "world_model_index":{"wmi":0.015,
   "should_ai_abstain":true,"thin_world":true},
 "macro_tilt":1.0,"alt_tilt":-1.0,
 "audit":{"evidence_ids":[
   "band:exchange:missing","band:macro:ready"]}}
```

Listing 2: 3/3-ready day still thin under schema ceiling.

```
{"date":"2026-01-16","asset":"ADA",
 "completeness":{"n_ready":3,"n_missing":0},
 "honesty":{"H":0.69,"U":0.38,"B_hier":0.35},
 "world_model_index":{"wmi":0.093,"acwmi":0.45,
   "should_ai_abstain":true,"thin_world":true},
 "band_status":{"exchange":"ready","macro":"ready",
   "alternative":"ready"}}
```

4.5 Four arms and frozen prompts

Actions ∈ {bullish, bearish, neutral, abstain}. **Compiled:** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true (and when the world is judged thin). **Ungated:** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness. **Raw:** mom5 only—thin price wiring for the ladder. **Blind:** "Today is d. How should I trade a?" with no feed. max WMI ≈ 0.093, so production Compiled abstains on every OOS day—a valve stress test under thin support. Compiled vs. Ungated differ only in enforcement (Listings 3–4); prompts, IS/OOS cut, and seeds are frozen. On thin Compiled days the evidence-alignment proxy (EAR) is 1.0 across vendors: rationales stay bound to disclosed `evidence_ids`.

Listing 3: Compiled (excerpt): hard valve.

```
1. If should_ai_abstain is true OR the world is thin,
   you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must match evidence_ids.
```

Listing 4: Ungated (excerpt): soft judgment.

```
1. Judge whether low WMI / missingness makes action
   unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why the hard valve is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the data end's job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.8). Ungated measures soft judgment from the same disclosed bundle; Raw measures fragment wiring; Blind measures no feed. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the valve.

4.6 Systems validation (engines)

Paper-lab smoke against the production engines returns a wired WMI/ACWMI sample with `should_ai_abstain` consistent with the configured index mode (WMI gate 0.2; ACWMI gate 0.35). The same indices appear in the consumer bundle; availability shocks O_t are queryable on the service surface.

5 Experiments

5.1 Setup

PIT panel ∼399 days × 10 liquid names (≈3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16).

Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

	Ready F1		Missing F1		Tilt-sign	
Model	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses n=2000; the no-LLM probe uses 199 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain rate (CE as avoided loss, not alpha); RQ3 ΔCE of the content rule vs. momentum.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility ≠ sufficiency chat. Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers "enough?" correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of "is this enough to trade?" is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) shows the valve contrast: Compiled abstains on `should_ai_abstain`; Raw goes bullish from mom5 = +1 alone.

Table 9: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

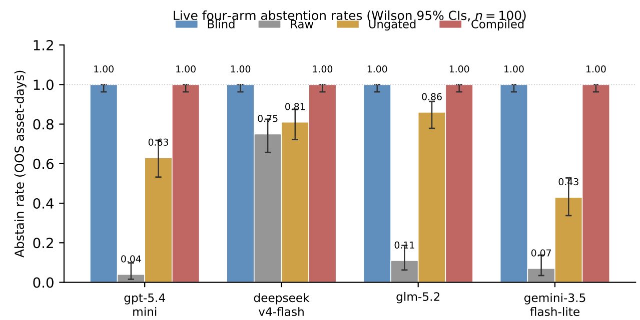

Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.

Table 10: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 9 and Figure 4: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean ΔCE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim.

Table 11: Full-OOS thin abstention (n=2000 asset-days; all thin under gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈0.56	–

*GLM coverage incomplete (Ungated n=1402; Raw n=675).

Table 12: Open slice (3/3 ready ≡ WMI ≥ 0.05; n≈1224). Field lower bound under soft judgment.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM n=398; Ungated GLM n=932; others n=1224. Bootstrap mean CE ≈+0.31 (B=5, 999 reps; one-sided p≈0.21).

Full-OOS replication. Table 11: on all 2000 OOS asset-days (every day thin under the 0.2 gate), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈0.56; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor): the failure mode that needs a data end is not "models never abstain," but "thin fragments make them act."

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek −3.12, GLM −4.28, Gemini −1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

Open slice. Table 12: on WMI ≥ 0.05 (≈61% OOS; n≈1224) Ungated keeps compiled content without the hard boolean—mean abs. ≈0.48, CE positive for every vendor (mean ≈+0.29; bootstrap ≈+0.31, CIs include 0). Production Compiled still abstains at 1.0; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). Point-positive open-slice CE and negative mixed-sample act-conditional CE can coexist: regime and act selection matter.

5.4 RQ3 (secondary): Handoff backtest—nonempty fields

The data end must be usable by a downstream decision rule without an LLM in the loop. Table 13 reads the same compiled panel under four handoff policies: production WMI gate 0.2, momentum on mom5, buy-and-hold, and a thick-ungated content rule on vintaged macro_tilt / alt_tilt. Production refuse is complete (Sharpe/CE 0, abstain 1.0); the content handoff is nonempty vs. momentum (Table 14, Figure 5): Sharpe 0.764 / CE +0.130 vs. 0.096 / −0.206; block-bootstrap ΔCE=0.334 (p=0.034). A 2017–2026 non-vintaged audit shows no edge; a date-scramble placebo flips CE from +0.132 to −0.254, supporting timed content.

Table 13: Handoff backtest on the compiled OOS panel (no LLM). Production gate vs. content rule.

Policy (reads bundle fields)	Sharpe	CE	Abstain
WMI gate 0.2 (production)	0.000	0.000	1.00
Momentum (mom5)	0.101	−0.202	0.00
Buy-and-hold	−1.399	−1.157	0.00
Thick-ungated tilts	**0.767**	**+0.132**	0.08

Table 14: No-LLM content contrast. Bootstrap ΔCE 0.334; raw gap 0.336.

Policy	Sharpe	CE
Buy-and-hold	−1.404	−1.163
Momentum	0.096	−0.206
World-bundle tilts	**0.764**	**+0.130**
Tilts − Momentum	–	0.334 (p=0.034)

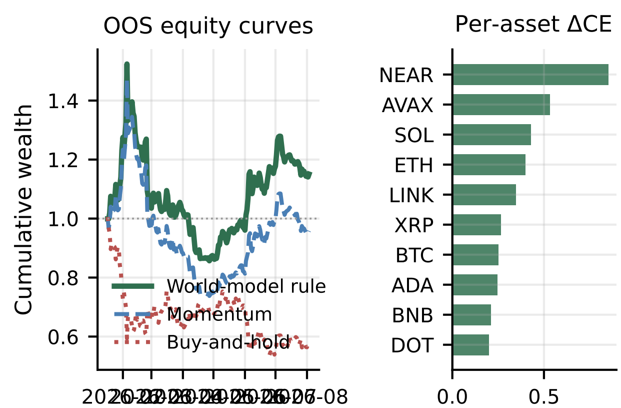

Figure 5: Secondary content probe: compiled tilts beat momentum; ΔCE positive for every asset.

Table 15: Content probe by archive density (OOS).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

*Block-bootstrap ΔCE; raw mean gap 0.336.

Density split and LOBO channel. Table 15: open/thick ΔCE=0.315; exchange-gap days abstain more (ΔCE=0.129). Table 16 and Figure 6 (Prop. 3.9): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).

Table 16: LOBO decomposition (ungated content probe). Content share 1.0.

Drop	ΔCE content	p	Content share
macro	−0.334	0.034	1.0
alternative	−0.334	0.034	1.0
exchange	0.000	1.000	—

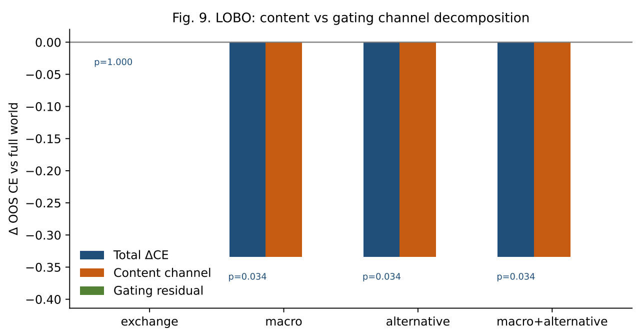

Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.

6 Discussion

What carries the paper. (i) The data end is legible: grounding ≈0.97, with sufficiency chat a poor substitute (RQ1 / Props. 3.6–3.7). (ii) Thin views must not drive action: soft text fails at 100-day and full-OOS scale; Blind refuses with no feed; Raw acts on fragments and loses; the hard valve works (RQ2 / Prop. 3.8). (iii) A no-LLM handoff shows production refuse vs. nonempty content on the same bundle (RQ3 / Prop. 3.9); LOBO content share 1.0. Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

Architecture. Semantics (Section 3) define the interfaces; Runtime (Section 4) implements Π_t and the service handoff; Experiments (Section 5) test seeing, gating, and nonempty fields in that order. Trading PnL is intentionally secondary: it confounds observation quality with policy skill.

Design. World bundle not feature dump; previous-close PIT with explicit freshness; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Service-layer health and bundle reads make the valve and the view callable by any downstream policy.

Reproducibility. Paper-lab smoke checks BandPIT/WMI/ACWMI wiring; PIT rebuild and the frozen four-arm harness regenerate the panel and live transcripts used here. Prompts, IS/OOS cut, seeds, and vendor IDs are fixed; credentials stay in environment variables. An anonymized harness ships on acceptance.

Scope. We evaluate the data end on the PIT archive above; decision and execution sit downstream.

7 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation layer: compile asynchronous evidence into a consumable PIT world bundle, disclose structural missingness, and hard-refuse when the view is too thin. Grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. Raw collapse; sufficiency chat alone fails; the hard valve abstains 1.0 on thin days (including full OOS) where soft judgment does not; vintaged tilts are nonempty (ΔCE=0.334; LOBO content share 1.0). Build the quant data end first: compile so models can see; refuse when they must not act.

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